

Parameter Estimation in Gaussian Process Regression for Deterministic Functions

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Table of contents

Introduction

Maximum likelihood estimation for deterministic f

What if f is a sample from a GP?

Two future results

Modelling Deterministic Functions

- Let $\Omega \subset \mathbb{R}^d$.
- Let $f: \Omega \rightarrow \mathbb{R}$ be a **deterministic** function.
- Suppose that f generates the **noiseless data**

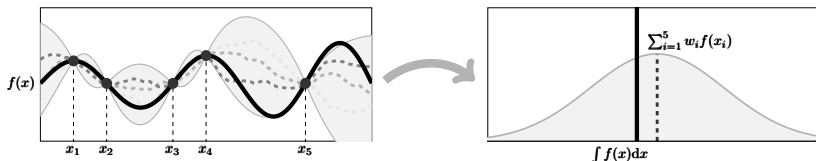
$$\mathcal{D}_N = \{(x_1, f(x_1)), \dots, (x_N, f(x_N))\} \text{ at some } x_i \in \Omega.$$

In this talk we model f using a **Gaussian process** f_{GP} .

- f need not be a sample from (or in any way related to) f_{GP} .
- The assumption that there is no noise is crucial.

Motivation

- *Probabilistic numerics.* Provide quantification of epistemic uncertainty arising from discretisation in numerical approximation.
- *Modelling of computer experiments.* Predict the output of a complex and computationally expensive piece of code.
- *Bayesian optimisation.* Construct surrogates to an objective function.



- (K2020) **Karvonen, Wynne, Tronarp, Oates & Särkkä (2020)**. Maximum likelihood estimation and uncertainty quantification for Gaussian process approximation of deterministic functions. *SIAM/ASA Journal on Uncertainty Quantification*, 8(3):926–958.
- (K2021a) **Karvonen (2021)**. Small sample spaces for Gaussian processes. *arXiv:2103.03169*.
- (K2021b) **Karvonen (2021)**. Estimation of the scale parameter for a misspecified Gaussian process model. *arXiv:2110.02810*.
- + Two papers in preparation.

Gaussian process interpolation I

- Let $K: \Omega \times \Omega \rightarrow \mathbb{R}$ be a **positive-definite covariance kernel**.
- Model f as a **Gaussian process** $f_{\text{GP}} \sim \text{GP}(0, K)$.
- Condition f_{GP} on the **data** $\mathcal{D}_N = \{(x_i, f(x_i))\}_{i=1}^N$.

The resulting conditional GP, $f_{\text{GP}} \mid \mathcal{D}_N$, has the **mean**

$$s_{f,N}(x) := \mathbb{E}[f_{\text{GP}}(x) \mid \mathcal{D}_N] = f_N^\top K_N^{-1} k_N(x) \quad (1)$$

and **variance**

$$P_N(x)^2 := \text{Var}[f_{\text{GP}}(x) \mid \mathcal{D}_N] = K(x, x) - k_N(x)^\top K_N^{-1} k_N(x), \quad (2)$$

where

$$(f_N)_i = f(x_i), \quad (k_N(x))_i = K(x, x_i) \quad \text{and} \quad (K_N)_{ij} = K(x_i, x_j).$$

Gaussian Process Interpolation II

- The conditional mean **interpolates** the data:

$$s_{f,N}(x_i) = f(x_i) \quad \text{for every } i = 1, \dots, N.$$

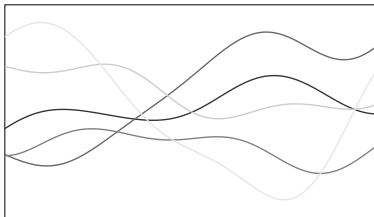
- The conditional variance **vanishes** at the data locations:

$$P_N(x_i)^2 = 0 \quad \text{for every } i = 1, \dots, N.$$

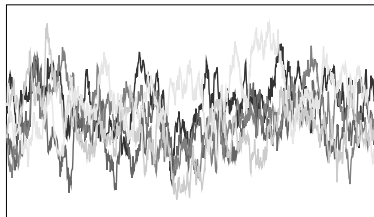
- The conditional variance **does not depend** on the data values $f(x_i)$.
- Properties of the kernel K define the properties of f_{GP} .

Gaussian process priors

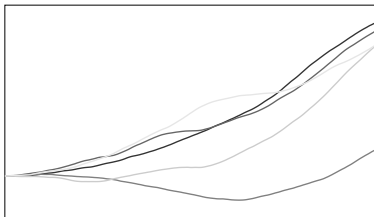
Gaussian: $K(x, y) = e^{-(x-y)^2/2}$



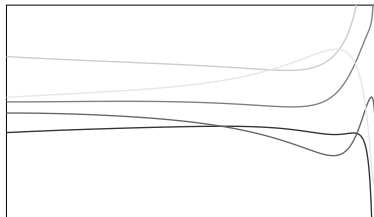
Matérn: $K(x, y) = e^{-|x-y|}$



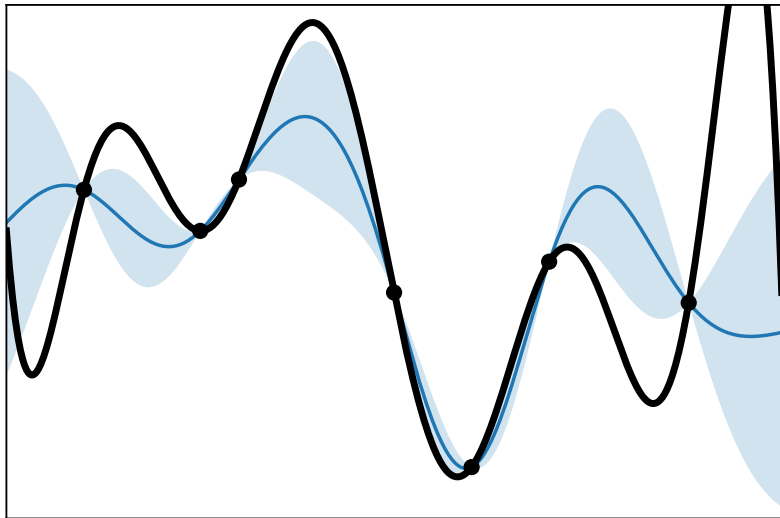
BM: $K(x, y) = \frac{\min\{x, y\}^3}{3} + \frac{|x-y|\min\{x, y\}^2}{2}$



Hardy: $K(x, y) = \frac{1}{1-xy}$



Gaussian Process Conditional



Objective

- Suppose that the prior covariance is parametric: $f_{\text{GP}} \sim \text{GP}(0, K_{\theta})$.
- The conditional process is

$$f_{\text{GP}} \mid \mathcal{D}_N \sim \text{GP}(s_{\theta, f, N}, P_{\theta, N}^2).$$

- For any $a \in (0, 1)$,

$$\mathbb{P}\left[|f_{\text{GP}}(x) - s_{\theta, f, N}(x)| \leq c(a)P_{\theta, N}(x, x) \mid \mathcal{D}_N\right] = 1 - a.$$

- Compute hyperparameter estimates $\theta(f, N)$ of θ .

Objective: Understand the behaviour of (i) $\theta(f, N)$ and (ii) the standard score

$$\frac{|f(x) - s_{\theta(f, N), f, N}(x)|}{P_{\theta(f, N), N}(x, x)} \quad \text{as } N \rightarrow \infty \quad (3)$$

for different f and hyperparameter estimation methods.

Table of contents

Introduction

Maximum likelihood estimation for deterministic f

What if f is a sample from a GP?

Two future results

Reproducing kernel Hilbert spaces

Reproducing kernel Hilbert space

Every covariance kernel $K: \Omega \times \Omega \rightarrow \mathbb{R}$ induces a unique *reproducing kernel Hilbert space* (RKHS) $\mathcal{H}(K)$ with inner product $\langle \cdot, \cdot \rangle_K$. The RKHS consists of functions $g: \Omega \rightarrow \mathbb{R}$ and the kernel has the *reproducing property*

$$\langle g, K(\cdot, x) \rangle_K = g(x) \quad \text{for any } g \in \mathcal{H}(K) \text{ and } x \in \Omega.$$

The GP conditional moments are related to optimal interpolation in $\mathcal{H}(K)$.

$s_{f,N}$ = minimum-norm interpolant

$$s_{f,N} = \arg \min_{s \in \mathcal{H}(K)} \{ \|s\|_K : s(x_i) = f(x_i) \text{ for every } i = 1, \dots, N \}$$

$P_N(x, x)$ = worst-case error

$$P_N(x, x) = \sup_{\|g\|_K \leq 1} |g(x) - s_{g,N}(x)|$$

Sobolev spaces

The **Sobolev space** $H^\alpha(\mathbb{R}^d)$ consists of functions $g \in L^2(\mathbb{R}^d)$ such that

$$\|g\|_\alpha^2 := \int_{\mathbb{R}^d} (1 + \|\xi\|^2)^\alpha |\widehat{g}(\xi)|^2 d\xi < \infty.$$

- For $\Omega \subset \mathbb{R}^d$ the space $H^\alpha(\Omega)$ is defined via restrictions.
- If $\alpha > n + d/2$ for $n \in \mathbb{N}$, then $H^\alpha(\mathbb{R}^d) \subset C^n(\mathbb{R}^d)$.

Sobolev kernel

Let $\alpha > d/2$ and $\Omega \subset \mathbb{R}^d$. A kernel $K: \Omega \times \Omega \rightarrow \mathbb{R}$ is a *Sobolev kernel of order α* if its RKHS $\mathcal{H}(K)$ is norm-equivalent ($\simeq$) to $H^\alpha(\Omega)$.

A Matérn kernel

$$K(x, y) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} \|x - y\|}{\lambda} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} \|x - y\|}{\lambda} \right)$$

of smoothness $\nu > 0$ is a Sobolev kernel of order $\nu + d/2$.

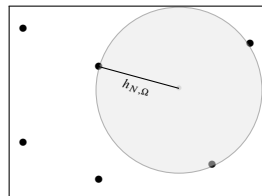
Sobolev rates

Suppose that

- $\Omega \subset \mathbb{R}^d$ is bounded and sufficiently regular (e.g., $\Omega = [0, 1]^d$).
- The points $x_1, \dots, x_N$ are **quasi-uniform**: the *fill-distance*

$$h_{N,\Omega} = \sup_{x \in \Omega} \min_{i=1, \dots, N} \|x - x_i\|$$

is of order $N^{-1/d}$ as $N \rightarrow \infty$.



Theorem (from approximation theory)

Let $d/2 < \beta \leq \alpha$. Suppose that $\mathcal{H}(K) \simeq H^\alpha(\Omega)$ and $f \in H^\beta(\Omega)$. Then

$$\sup_{x \in \Omega} |f(x) - s_{f,N}(x)| \leq C_1 \|f\|_\beta N^{-\beta/d+1/2} \quad (4)$$

and

$$C_2 N^{-\alpha/d+1/2} \leq P_N(x) \leq C_3 N^{-\alpha/d+1/2} \quad \text{for } x \notin \{x_i\}_{i=1}^\infty. \quad (5)$$

Why parameter estimation is necessary

If K is a Sobolev kernel of order α and $f \in H^\beta(\Omega)$ for $\beta \leq \alpha$, then

$$\frac{|f(x) - s_{f,N}(x)|}{P_N(x, x)} \leq \frac{C_1 \|f\|_\beta N^{-\beta/d+1/2}}{C_2 N^{-\alpha/d+1/2}} = C_4 N^{(\alpha-\beta)/d}.$$

$\Rightarrow$ If $f \notin \mathcal{H}(K) \simeq H^\alpha(\Omega)$, it may be necessary estimate kernel parameters in order to prevent overconfidence:

$$\frac{\alpha - \beta}{d} > 0.$$

Maximum likelihood estimation

The log-likelihood function is

$$\ell(\theta) = -\frac{1}{2} \left[f_N^\top K_{\theta,N}^{-1} f + \log \det K_{\theta,N} + N \log(2\pi) \right],$$

where $(f_N)_i = f(x_i)$ and $(K_{\theta,N})_{ij} = K_\theta(x_i, x_j)$.

Fix K and use the simple parametrisation $K_\sigma = \sigma^2 K$ for $\sigma \geq 0$. Then

$$s_{\sigma,f,N}(x) = s_{f,N}(x) \quad \text{and} \quad P_{\sigma,N}(x, y) = \sigma P_N(x, y).$$

$$\sigma_{\text{ML}}(f, N) = \arg \max_{\sigma \geq 0} \ell(\sigma) = \sqrt{\frac{f_N^\top K_N^{-1} f_N}{N}} = \frac{\|s_{f,N}\|_K}{\sqrt{N}} \quad (6)$$

and

$$\text{standard score} = \frac{|f(x) - s_{f,N}(x)|}{\sigma_{\text{ML}}(f, N) P_N(x, x)} \quad (7)$$

First results — $f \in \mathcal{H}(K)$

Proposition (Proposition 3.1 in K2020)

If $f \in \mathcal{H}(K)$ and $f(x_i) \neq 0$ for some x_i , then there is $c > 0$ such that

$$c N^{-1/2} \leq \sigma_{\text{ML}}(f, N) \leq \|f\|_K N^{-1/2}. \quad (8)$$

Theorem (Theorem 3.2 in K2020)

If $f \in \mathcal{H}(K)$ and $f(x_i) \neq 0$ for some x_i , then there is $C > 0$ such that

$$\frac{|f(x) - s_{f,N}(x)|}{\sigma_{\text{ML}}(f, N) P_N(x, x)} \leq C \sqrt{N}. \quad (9)$$

At most “slow” (i.e., $\sqrt{N}$) overconfidence if $f \in \mathcal{H}(K)$.

Second result — $f \in H^\beta(\Omega)$ and $\mathcal{H}(K) \simeq H^\alpha(\Omega)$

Let $\beta \in (d/2, \alpha]$. Suppose that Ω is regular and $\{x_i\}_{i=1}^\infty$ are quasi-uniform.

Proposition (Proposition 4.5 in K2020)

If $\mathcal{H}(K) \simeq H^\alpha(\Omega)$ and $f \in H^\beta(\Omega)$, then

$$\sigma_{\text{ML}}(f, N) \leq C_1 N^{(\alpha-\beta)/d-1/2} \|f\|_{H^\beta(\Omega)}. \quad (10)$$

$\implies$ the behaviour of $\sigma_{\text{ML}}(f, N)$ tells how “far” from $\mathcal{H}(K)$ the function f is.

Theorem (Theorem 4.10 in K2020)

If $\mathcal{H}(K) \simeq H^\alpha(\Omega)$ and f has “exact smoothness” β for $\lfloor \beta \rfloor > d/2$, then

$$\frac{|f(x) - s_{f,N}(x)|}{\sigma_{\text{ML}}(f, N) P_N(x, x)} \leq C_2(f) (\log N)^{\alpha/(2\beta)} \sqrt{N}. \quad (11)$$

At most “slow” (i.e., $\approx \sqrt{N}$) overconfidence if f is rougher than $\mathcal{H}(K)$.

Some implications

- Overconfidence cannot be ruled out, but at least it cannot be overly severe: the standard score is approximately $O(N^{1/2})$.
- Simple maximum likelihood estimation of a scaling parameter provides strong protection against smoothness misspecification.
- MLE does not detect undersmoothing by the model:

$$\sigma_{\text{ML}}(f, N) \asymp N^{-1/2} \quad \text{for every non-zero } f \in \mathcal{H}(K).$$

- It can be shown (**Theorem 4.11 in K2020**) that underconfidence occurs if there is (roughly speaking) sufficient undersmoothing:

$$\mathcal{H}(K) \simeq H^{\alpha}(\Omega) \quad \text{and} \quad f \in H^{2\alpha}(\Omega).$$

Numerical results: confidence intervals

$\mathcal{H}(K) \simeq H^2([0, 1])$ and $f \in H^\beta([0, 1])$ for $\beta = \frac{2}{2}, \frac{3}{2}, \dots, \frac{6}{2}$.

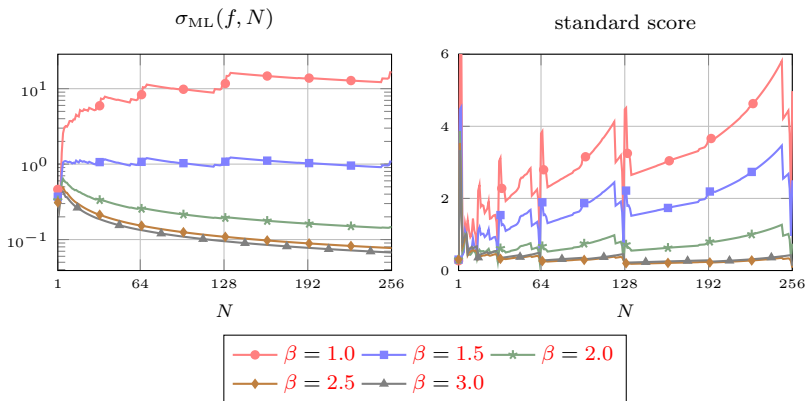


Table of contents

Introduction

Maximum likelihood estimation for deterministic f

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Two future results

Sample path properties

Theorem (e.g., Steinwart 2019)

Let $f_{\text{GP}} \sim \text{GP}(0, K)$ and suppose that K is a Sobolev kernel of order $\alpha > d/2$. Then

$$\mathbb{P}[f_{\text{GP}} \in H^{\beta}(\Omega)] = 0 \quad \text{if} \quad \beta \geq \alpha - d/2$$

and

$$\mathbb{P}[f_{\text{GP}} \in H^{\beta}(\Omega)] = 1 \quad \text{if} \quad \beta < \alpha - d/2.$$

The samples of the Gaussian process f_{GP} are therefore “ $d/2$ less smooth” than the RKHS $\mathcal{H}(K)$.

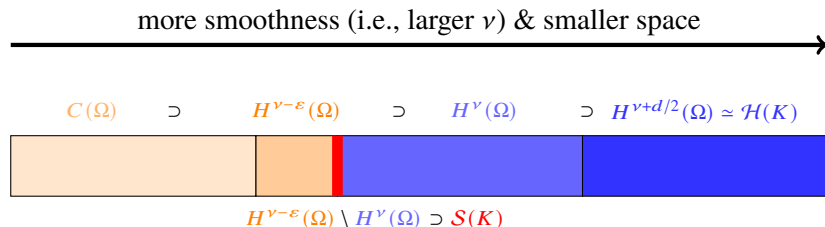
$\implies$ Samples are *not* in the RKHS $\mathcal{H}(K) \simeq H^{\alpha}(\Omega)$!

$\implies$ Samples have “exact smoothness” $\alpha - d/2$ if $\mathcal{H}(K) \simeq H^{\alpha}(\Omega)$.

Matérn kernels

Let K be a Matérn kernel of smoothness $\nu > 0$ and $\Omega \subset \mathbb{R}^d$ sufficiently regular. Then

- $\mathcal{H}(K)$ is norm-equivalent to the Sobolev space $H^{\nu+d/2}(\Omega)$.
- $\mathbb{P}[f_{\text{GP}} \in H^{\nu-\varepsilon}(\Omega)] = 1$ if and only if $\varepsilon > 0$.



The samples of f_{GP} can be thought of as elements of the “Sobolev slice”

$$\mathcal{S}(K) \subset H^{\nu-\varepsilon}(\Omega) \setminus H^{\nu}(\Omega) \quad \text{for every } \varepsilon > 0.$$

Expected scale MLE for samples

Suppose that the data-generating function is a Gaussian process

$$f = f_{\text{GP}}^* \sim \text{GP}(0, R)$$

but the model is $f_{\text{GP}} \sim \text{GP}(0, K)$. Then

$$\mathbb{E}_{f_{\text{GP}}^*} [\sigma_{\text{ML}}(f_{\text{GP}}^*, N)^2] = \mathbb{E}_{f_{\text{GP}}^*} \left[\frac{(f_{\text{GP},N}^*)^\top K_N^{-1} f_{\text{GP},N}^*}{N} \right] = \frac{\text{trace}(R_N K_N^{-1})}{N}.$$

Theorem (Theorem 4.2 in K2021b)

Suppose that $\mathcal{H}(K) \simeq H^\alpha([0, 1]^d)$ and $\mathcal{H}(R) \simeq H^{\alpha_0}([0, 1]^d)$ for $\alpha \geq \alpha_0 > d/2$. If the points $\{x_i\}_{i=1}^\infty$ are quasi-uniform, then

$$C_1 N^{2(\alpha - \alpha_0)/d} \leq \mathbb{E}_{f_{\text{GP}}^*} [\sigma_{\text{ML}}(f_{\text{GP}}^*, N)^2] \leq C_2 N^{2(\alpha - \alpha_0)/d}.$$

Comparison to the deterministic case

Suppose that $\mathcal{H}(K) \simeq H^\alpha([0, 1]^d)$ and that $\{x_i\}_{i=1}^\infty$ are quasi-uniform.

Recall that the samples “have smoothness $\alpha - d/2$ ”.

Deterministic

Let $f \in H^{\alpha-d/2}([0, 1]^d)$ so that f *does not have less smoothness* than the samples of $f_{\text{GP}} \sim \text{GP}(0, K)$. For $\beta = \alpha - d/2$ we get

$$\sigma_{\text{ML}}(f, N)^2 \leq C_1 N^{2(\alpha-\beta)/d-1} \|f\|_{H^\beta(\Omega)}^2 = C_1 \|f\|_{H^{\alpha-d/2}(\Omega)}^2. \quad (12)$$

Stochastic

Let $f = f_{\text{GP}}^* \sim \text{GP}(0, R)$ for R such that $\mathcal{H}(R) \simeq H^\alpha([0, 1]^d)$. Then $f_{\text{GP}} \sim \text{GP}(0, K)$ and f_{GP}^* *have similar paths*. For $\alpha_0 = \alpha$ we get

$$\mathbb{E}_{f_{\text{GP}}^*} [\sigma_{\text{ML}}(f_{\text{GP}}^*, N)^2] \asymp N^{(\alpha-\alpha_0)/d} = 1. \quad (13)$$

Table of contents

Introduction

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Is cross-validation better than MLE?

The **leave-one-out cross-validated** estimate of σ is

$$\sigma_{\text{cv}}(f, N) = \frac{1}{N} \sum_{i=1}^N \left(\frac{f(x_i) - s_{f, N \setminus i}(x_i)}{P_{N \setminus i}(x_i, x_i)} \right)^2,$$

where $s_{f, N \setminus i}$ and $P_{N \setminus i}$ are the GP conditional mean and std based on the data set $\mathcal{D}_N \setminus \{(x_i, f(x_i))\}$.

- Recall that for any non-zero $f \in \mathcal{H}(K)$ we have

$$\sigma_{\text{ML}}(f, N) \asymp N^{-1/2} \quad \text{for any non-zero } f \in \mathcal{H}(K).$$

- At least in some cases it can be proved¹ that the rate of decay of

$$\sigma_{\text{cv}}(f, N) \quad \text{depends on} \quad f \in \mathcal{H}(K).$$

$\implies$ Cross-validation is **more sensitive** to the smoothness of f than MLE.

¹Work in progress with M. Naslidnyk, M. Mahsereci and M. Kanagawa.

Estimating the lengthscale parameter

The kernel K is stationary if it can be written as

$$K(x, y) = \Phi\left(\frac{x - y}{\lambda}\right) \quad \text{for some} \quad \Phi: \mathbb{R}^d \rightarrow \mathbb{R},$$

where $\lambda > 0$ is the **lengthscale** parameter.

Theorem (to appear in a paper with C. Oates)

Let $N \geq 2$ and suppose that $\mathcal{H}(K) \simeq H^\alpha(\mathbb{R}^d)$ for some $\alpha > d/2$. If the data vector is **constant**,

$$f_N = (c, \dots, c) \in \mathbb{R}^N \quad \text{for some} \quad c \in \mathbb{R},$$

then

$$\lambda_{\text{ML}} = \infty.$$

Conclusion

- Simple MLE of the scaling parameter σ in $K_\sigma(x, y) = \sigma^2 K(x, y)$ provides **protection against misspecification**.
- **Overconfidence is possible** but at least it cannot be **too severe**.
- Samples from a GP are elements of a “small” set of functions. This is manifested in the sample results being “nicer”.
- Cross-validation may be **more sensitive** to the smoothness of f than MLE.

Additional references

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Thank you for your attention!